

Appendix: Theoretical Analysis of the Augmented Synthetic Historical Control Estimator (ASHC)

Setup and Assumptions

For the purposes of self-containment, let $\mathcal{T} = \{1, 2, \dots, T\}$ index time at a monthly frequency. Define the pre-treatment period as $\mathcal{T}_1 := \{t \in \mathcal{T} : t \leq T_0\}$ and the post-treatment period as $\mathcal{T}_2 := \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$. Let $m \in \mathbb{N}$ denote the evaluation window length. Define the evaluation period as the final m months of the pre-treatment period $\mathcal{T}_{\text{eval}} := \{T_0 - m + 1, \dots, T_0\} \subset \mathcal{T}_1$.

Let $\mathbf{y} = (y_1, y_2, \dots, y_T)^\top \in \mathbb{R}^T$ denote the observed outcome vector for the treated unit. Define the pre-treatment outcome vector $\mathbf{y}_{\text{pre}} := \mathbf{y}_{\mathcal{T}_1} \in \mathbb{R}^{T_0}$, the evaluation subvector $\mathbf{y}_{\text{eval}} := \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector $\mathbf{y}_{\text{post}} := \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$.

The evaluation subvector $\mathbf{y}_{\text{eval}}$ is the target pattern to be reconstructed using convex combinations of earlier segments from the pre-treatment period.

SHC by [Chen et al. \[2024\]](#) constructs

$$N = T_0 - m - n + 1$$

overlapping temporal segments of length m . Each segment $\tilde{\mathbf{y}}_{[i, i+m-1]}$ is a contiguous subvector of the smoothed pre-treatment series. These comprise the donor matrix

$$\tilde{\mathbf{Y}}_{\text{pre}} := [\tilde{\mathbf{y}}_{[1, m]} \quad \tilde{\mathbf{y}}_{[2, m+1]} \quad \dots \quad \tilde{\mathbf{y}}_{[N, N+m-1]}] \in \mathbb{R}^{m \times N}.$$

Example 0.1. Suppose we have $T_0 = 60$ months of pre-treatment data and we wish to construct donor segments of length $m = 12$ months to predict a post-treatment horizon of $n = 6$ months. The number of eligible segments is:

$$N = 60 - 12 - 6 + 1 = 43.$$

Each donor segment is a vector of 12 consecutive smoothed pre-treatment outcomes.

In the third step, we solve the following program to obtain the optimal weights:

$$\begin{aligned} \mathbf{w}^* = \underset{\mathbf{w} \in \mathbb{R}^N}{\text{argmin}} \quad & \left\| \tilde{\mathbf{y}}_{\text{eval}} - \tilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \\ \text{subject to} \quad & \mathbf{w} \geq \mathbf{0}, \quad \mathbf{1}^\top \mathbf{w} = 1. \end{aligned}$$

Let $\mathbf{w}^{\text{SHC}}$ denote the SHC solution, and $\mathbf{w}^{\text{ASHC}}$ the ASHC solution:

$$\mathbf{w}^{\text{ASHC}} = \arg \min_{\mathbf{w} \in \mathbb{R}^N} \left\{ \frac{1}{2\lambda} \left\| \tilde{\mathbf{y}} - \tilde{\mathbf{Y}} \mathbf{w} \right\|^2 + \left\| \mathbf{w} - \mathbf{w}^{\text{SHC}} \right\|^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \right\} \quad \text{s.t.} \quad \sum_{i=1}^N w_i = 1,$$

where $\lambda > 0$ is the regularization parameter, $\varsigma > 0$ is the penalty parameter for the low-variance directions, and $\mathbf{C}_0 \in \mathbb{R}^{N \times k}$ contains the eigenvectors of the Gram matrix $\mathbf{G} = \tilde{\mathbf{Y}}^\top \tilde{\mathbf{Y}}$ corresponding to eigenvalues less than a threshold $\tau > 0$.

Assumptions

Assumption 0.1 (Linearity with noise). *There exists $\mathbf{w}^* \in \mathbb{R}^N$ such that*

$$\tilde{\mathbf{y}} = \tilde{\mathbf{Y}}\mathbf{w}^* + \boldsymbol{\eta},$$

where $\boldsymbol{\eta} \in \mathbb{R}^m$ is a mean-zero noise vector.

Assumption 0.2 (Bounded noise). $\|\boldsymbol{\eta}\| \leq \delta$ for some $\delta > 0$.

Assumption 0.3 (Smoothness of latent trend). *Let $\ell(t)$ denote the latent smooth component of the treated potential outcome. Suppose $\ell(t)$ is $H + 1$ times differentiable with $|\ell^{(H+1)}(t)| < \epsilon$ for all t and some $H \geq 3$.*

Assumption 0.4 (Operator norm bound). $\|\tilde{\mathbf{Y}}\| \leq \gamma$.

Assumption 0.5 (Simplex feasibility). $\mathbf{w}^{SHC} \in \Delta^{N-1} := \{\mathbf{w} \in \mathbb{R}^N : \sum w_i = 1, w_i \geq 0\}$.

Lemma 0.1 (Deviation from SHC under Affine Constraint). *Let $\Delta := \mathbf{w}^{ASHC} - \mathbf{w}^{SHC}$. Under Assumptions (A1)–(A5), we have*

$$\|\Delta\| \leq \frac{\epsilon_{SHC}}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}},$$

where $\epsilon_{SHC} := \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{SHC}\|$ and $\|\mathbf{C}_0\|^2 := \lambda_{\max}(\mathbf{C}_0^\top \mathbf{C}_0)$.

Proof. To show how much the ASHC may deviate in sample, we can begin by rewriting our objective function which minimizes

$$J(\mathbf{w}) = \frac{1}{2\lambda} \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}\|^2 + \|\mathbf{w} - \mathbf{w}^{SHC}\|^2 + \varsigma \|\mathbf{C}_0^\top \mathbf{w}\|_2^2 \quad \text{s.t. } \mathbf{1}^\top \mathbf{w} = 1$$

Explanation: The first term is a fit term. It encourages the solution to have the ASHC weights fit closely to the evaluation period outcome vector. The second term penalizes deviation from the SHC weights, or, allows us to extrapolate beyond the convex hull of donor segments. The third term, as also discussed by [Chen et al. \[2024\]](#), stabilizes the solution in directions of low variance.

If we ignored the affine constraint and simply minimized $J(\mathbf{w})$, we would obtain an unconstrained solution $\mathbf{w}^{\text{uncon}}$ that may not satisfy $\mathbf{1}^\top \mathbf{w} = 1$, and therefore would not be a valid ASHC weight vector. Thus, we need to allow ASHC to have a feasible solution set. to do this, we use the Lagrangian, where we form

$$\mathcal{L}(\mathbf{w}, \mu) = J(\mathbf{w}) + \mu(\mathbf{1}^\top \mathbf{w} - 1),$$

with $\mu \in \mathbb{R}$ being a Lagrange multiplier. The Lagrangian corrects this by introducing μ , which acts like a "force" that pins the solution back onto the hyperplane of feasible weights. Importantly, while μ shifts the right-hand side of the optimality condition, it does not affect the curvature matrix M that controls $\|\Delta\|$. Thus feasibility is enforced without disrupting the conditioning argument.

Step 3. Differentiate each term.

$$\begin{aligned} \nabla_{\mathbf{w}} \frac{1}{2\lambda} \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}\|^2 &= \frac{1}{\lambda} \tilde{\mathbf{Y}}^\top (\tilde{\mathbf{Y}}\mathbf{w} - \tilde{\mathbf{y}}), \\ \nabla_{\mathbf{w}} \|\mathbf{w} - \mathbf{w}^{SHC}\|^2 &= 2(\mathbf{w} - \mathbf{w}^{SHC}), \\ \nabla_{\mathbf{w}} \varsigma \|\mathbf{C}_0^\top \mathbf{w}\|^2 &= 2\varsigma \mathbf{C}_0 \mathbf{C}_0^\top \mathbf{w}, \\ \nabla_{\mathbf{w}} [\mu(\mathbf{1}^\top \mathbf{w} - 1)] &= \mu \mathbf{1}. \end{aligned}$$

Explanation: Each gradient follows from standard rules: quadratic forms give $2A^\top(Aw - b)$, while $\mu(\mathbf{1}^\top w - 1)$ differentiates to $\mu \mathbf{1}$.

Step 4. Stationarity condition. Setting $\nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}, \mu) = 0$:

$$\left(\frac{1}{\lambda} \tilde{\mathbf{Y}}^\top \tilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma \mathbf{C}_0 \mathbf{C}_0^\top \right) \mathbf{w} = \frac{1}{\lambda} \tilde{\mathbf{Y}}^\top \tilde{\mathbf{y}} + 2\mathbf{w}^{SHC} - \mu \mathbf{1}.$$

Explanation: This is the first-order condition: the gradient must vanish. The $\mu \mathbf{1}$ term shifts the right-hand side to enforce feasibility.

Step 5. Deviation form. Define

$$M := \frac{1}{\lambda} \tilde{\mathbf{Y}}^\top \tilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma \mathbf{C}_0 \mathbf{C}_0^\top.$$

Letting $\Delta = \mathbf{w}^{\text{ASHC}} - \mathbf{w}^{\text{SHC}}$, we subtract the equation at $\mathbf{w}^{\text{SHC}}$ to obtain

$$M\Delta = \frac{1}{\lambda} \tilde{\mathbf{Y}}^\top (\tilde{\mathbf{y}} - \tilde{\mathbf{Y}} \mathbf{w}^{\text{SHC}}) - \mu \mathbf{1}.$$

Explanation: This isolates the deviation Δ , showing it depends on the SHC residual and the Lagrange term.

Step 6. Handle the μ term. The affine constraint implies $\mathbf{1}^\top \Delta = 0$, so Δ is orthogonal to $\mathbf{1}$. Thus the $\mu \mathbf{1}$ term, although present in the equation, does not affect Δ 's magnitude in the feasible subspace.

Explanation: Geometrically, $\mu \mathbf{1}$ is parallel to the constraint direction, while Δ lives in the orthogonal hyperplane. So μ enforces feasibility but does not change the conditioning that controls $\|\Delta\|$.

Step 7. Bound Δ . Let $r := \tilde{\mathbf{y}} - \tilde{\mathbf{Y}} \mathbf{w}^{\text{SHC}}$, with $\|r\| = \epsilon_{\text{SHC}}$. Then

$$\|M\Delta\| \leq \frac{1}{\lambda} \|\tilde{\mathbf{Y}}^\top r\| \leq \frac{\gamma}{\lambda} \epsilon_{\text{SHC}},$$

using Assumption (A4).

Because the smallest eigenvalue of M is at least $2 + 2\varsigma \|\mathbf{C}_0\|^2$, we conclude

$$\|\Delta\| \leq \frac{\gamma}{2\lambda + 2\varsigma \|\mathbf{C}_0\|^2} \epsilon_{\text{SHC}}.$$

Explanation: Strong convexity of M guarantees $\|M^{-1}\|$ is bounded. Thus the deviation is controlled by the SHC residual, scaled down by the effective regularization strength. $\square$

Remark 0.1. The key role of the affine constraint $\mathbf{1}^\top \mathbf{w} = 1$ is to introduce a Lagrange multiplier μ , which shifts the optimality condition by a vector in the span of $\mathbf{1}$. Because Δ must satisfy $\mathbf{1}^\top \Delta = 0$, this shift does not affect the conditioning of M and thus does not alter the deviation bound. The effective regularization is governed entirely by the spectrum of M .

Lemma 0.2. Under (A4), the error from deviating off SHC weights satisfies

$$\|\tilde{\mathbf{Y}}\Delta\| \leq \gamma \|\Delta\|.$$

Remark 0.2. Lemma 0.2 ensures that deviations of the ASHC weights from the SHC weights, measured in ℓ_2 norm, translate into controlled deviations in the predicted outcomes. The operator norm bound (A4) prevents $\tilde{\mathbf{Y}}$ from amplifying small changes in the weights disproportionately.

Lemma 0.3. Under (A3), for horizon $k > 0$,

$$|\ell(T_0 + k) - \ell^{\text{SHC}}(T_0 + k)| \leq b_\epsilon(H, k) := \frac{2\epsilon |k|^{H+1}}{(H+1)!}.$$

Proof. Let $\ell(t)$ denote the latent smooth component of the treated outcome. By (A3), $\ell(t)$ is $(H+1)$ times differentiable with $|\ell^{(H+1)}(t)| \leq \epsilon$.

Step 1. Taylor expansion around T_0 . By Taylor's theorem,

$$\ell(T_0 + k) = \sum_{h=0}^H \frac{\ell^{(h)}(T_0)}{h!} k^h + R_{H+1}(k),$$

with remainder

$$R_{H+1}(k) = \frac{\ell^{(H+1)}(\xi)}{(H+1)!} k^{H+1}, \quad \text{for some } \xi \in [T_0, T_0 + k].$$

Step 2. Bound the remainder.

$$|R_{H+1}(k)| \leq \frac{\epsilon}{(H+1)!} |k|^{H+1}.$$

Step 3. SHC approximation of lower-order terms. By construction, SHC matches the first H terms of this expansion up to error of the same order as the remainder.

Step 4. Combine bounds.

$$|\ell(T_0 + k) - \ell^{\text{SHC}}(T_0 + k)| \leq \frac{2\epsilon}{(H+1)!} |k|^{H+1}.$$

□

Remark 0.3. Lemma 0.3 formalizes the SHC bias as a Taylor remainder. The SHC approximation matches derivatives of $\ell(t)$ at T_0 up to order H , leaving bias decaying like $|k|^{H+1}$. Thus, for smooth latent trends and moderate horizons, SHC bias is negligible.

Main Result

Proposition 0.1. Under assumptions (A1)–(A5), the ASHC prediction error is bounded by

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\| \leq \epsilon_{\text{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}} \right) + b_\epsilon(H, k) + \delta.$$

Proof. We want to bound the prediction error of the ASHC estimator,

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\|.$$

Step 1: Decompose the error using the triangle inequality. Let $\mathbf{w}^*$ denote the ideal weights from the SHC estimator without the ASHC penalty. Define the difference $\Delta := \mathbf{w}^{\text{ASHC}} - \mathbf{w}^*$. Then,

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\| = \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^* - \tilde{\mathbf{Y}}\Delta\| \leq \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^*\| + \|\tilde{\mathbf{Y}}\Delta\|.$$

The first term, $\epsilon_{\text{SHC}} := \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^*\|$, is the SHC prediction error without augmentation.

Step 2: Control the second term using operator norm. By Lemma 0.2, the matrix $\tilde{\mathbf{Y}}$ satisfies

$$\|\tilde{\mathbf{Y}}\Delta\| \leq \gamma\|\Delta\|,$$

where γ is an operator norm bound capturing how the weights perturbations translate into prediction changes.

Step 3: Bound the weight difference $\|\Delta\|$. From Lemma 1 (stated earlier), the ASHC weight vector $\mathbf{w}^{\text{ASHC}}$ stays close to the SHC weights in Euclidean norm:

$$\|\Delta\| = \|\mathbf{w}^{\text{ASHC}} - \mathbf{w}^*\| \leq \frac{\epsilon_{\text{SHC}}}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}},$$

where the denominator reflects the strength of the ASHC penalty, ensuring stability.

Step 4: Incorporate smoothness bias and noise. Finally, the total prediction error also includes the smoothness bias $b_\epsilon(H, k)$ due to approximating the latent function by a truncated Taylor expansion of order H , and a noise term bounded by δ (from assumption (A2)).

Putting all these together, we get

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\| \leq \epsilon_{\text{SHC}} + \gamma\|\Delta\| + b_\epsilon(H, k) + \delta \leq \epsilon_{\text{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H, k) + \delta,$$

which completes the proof. $\square$

Discussion

Proposition~0.1 decomposes ASHC error into: 1. SHC fit error (ϵ_{SHC}), 2. amplification from ASHC deviation, 3. smoothness bias $b_\epsilon(H, k)$, 4. noise δ .

The penalty term $\varsigma\|\mathbf{C}_0^\top \mathbf{w}\|_2^2$ reduces deviation but may bias if $\mathbf{w}^*$ has components in $\mathbf{C}_0$ directions. The trade-off between λ and ς governs stability vs. fidelity.

Asymptotic Results

Theorem 0.1. *Under (A1)–(A5), as $m \rightarrow \infty$ and $H \rightarrow \infty$, for fixed $k > 0$,*

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\| \rightarrow \delta.$$

If $\boldsymbol{\eta}$ consists of independent mean-zero sub-Gaussian entries, then

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\| \xrightarrow{p} 0.$$

Proof. Starting from Proposition 0.1, we have

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\| \leq \epsilon_{\text{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H, k) + \delta.$$

Step 1. Behavior of ϵ_{SHC} . By (A3), $\ell(t)$ is $(H+1)$ -times differentiable. Gaussian kernel smoothing in SHC ensures

$$\epsilon_{\text{SHC}} \rightarrow 0 \quad \text{as } m \rightarrow \infty, H \rightarrow \infty.$$

[Chen et al. \[2024\]](#)

Step 2. Behavior of ASHC penalty term. Since $\epsilon_{\text{SHC}} \rightarrow 0$, the inflation factor

$$\frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}} \epsilon_{\text{SHC}}$$

vanishes asymptotically.

Step 3. Behavior of smoothness bias $b_\epsilon(H, k)$. From Taylor's theorem,

$$\ell(T_0 + k) = \sum_{h=0}^H \frac{\ell^{(h)}(T_0)}{h!} k^h + R_{H+1}(k),$$

with

$$|R_{H+1}(k)| \leq \frac{\epsilon}{(H+1)!} |k|^{H+1}.$$

Thus

$$b_\epsilon(H, k) \leq \frac{2\epsilon}{(H+1)!} |k|^{H+1} \rightarrow 0 \quad \text{as } H \rightarrow \infty.$$

Step 4. Behavior of noise term. By (A2), $\|\boldsymbol{\eta}\| \leq \delta$. If $\boldsymbol{\eta}$ is i.i.d. sub-Gaussian with variance proxy σ^2 , then

$$\frac{\|\boldsymbol{\eta}\|}{m} \xrightarrow{p} 0.$$

Step 5. Combine bounds. Therefore,

$$\limsup_{m \rightarrow \infty} \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{\text{ASHC}}\| \leq \delta.$$

With sub-Gaussian noise, the limit is 0 in probability. $\square$

Corollary 0.1. *Under (A1)–(A5), the ASHC estimator is asymptotically unbiased up to δ . If $\boldsymbol{\eta}$ is i.i.d. mean-zero sub-Gaussian with variance proxy σ^2 and bandwidth $h_m \asymp m^{-1/(2H+3)}$, then as $m \rightarrow \infty$,*

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{\text{ASHC}}\| = O_p\left(m^{-H/(2H+3)}\right).$$

For large H , this approaches the parametric rate $O_p(m^{-1/2})$.

Proof. Part 1: Consistency. From Theorem 0.1, the error converges to δ , or 0 in probability under sub-Gaussian noise.

Part 2: Convergence Rate. From Proposition 0.1, we have

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{\text{ASHC}}\| \leq \epsilon_{\text{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H, k) + \delta.$$

Step 1. Bias term. From Lemma 0.3,

$$b_\epsilon(H, k) = O(h_m^H).$$

Step 2. Stochastic error term. With sub-Gaussian noise, the smoothed variance is

$$O_p((mh_m)^{-1/2}).$$

Step 3. SHC fit error. Combining,

$$\epsilon_{\text{SHC}} = O_p(h_m^H + (mh_m)^{-1/2}).$$

Step 4. Optimal bandwidth. Equating orders $h_m^H \asymp (mh_m)^{-1/2}$ yields

$$h_m \asymp m^{-1/(2H+3)}.$$

Step 5. Convergence rate. Substituting,

$$\epsilon_{\text{SHC}} = O_p(m^{-H/(2H+3)}).$$

Thus the total error is

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{\text{ASHC}}\| = O_p(m^{-H/(2H+3)}).$$

Step 6. Near-parametric rate. As $H \rightarrow \infty$,

$$\frac{H}{2H+3} \rightarrow \frac{1}{2}.$$

So the rate approaches $O_p(m^{-1/2})$. $\square$

Remark 0.4. *The deterministic noise bound δ in (A2) corresponds to $\|\boldsymbol{\eta}\| \leq \delta$. Under sub-Gaussian noise, $\delta = O(\sigma\sqrt{m})$, while kernel smoothing reduces effective noise to $O_p((mh_m)^{-1/2})$. This explains the appearance of the nonparametric rate.*

References

Yi-Ting Chen, Jui-Chung Yang, and Tzu-Ting Yang. Synthetic historical control for policy evaluation. Available at SSRN: <https://ssrn.com/abstract=4995085>, September 2024. URL <http://dx.doi.org/10.2139/ssrn.4995085>.